



BOOK OF ABSTRACTS

6TH INTERNATIONAL MULTIDISCIPLINARY
CONFERENCE FOR YOUNG RESEARCHERS

*"Science and Innovation: Advancing
the Path to a Sustainable Future"*

9-10 October 2025
Prague, Czech Republic

ISBN: 978-80-213-3513-4

Publisher: Czech University of Life Sciences Prague



Stakeholder Dynamics and Decision Pathways in Farmers' Transition from Rice to Vegetables in Indonesia

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Abstract: In Indonesia's highland regions, where conditions favor horticulture, smallholder farmers are increasingly encouraged to shift from local rice cultivation to high-value vegetable production. This transition offers benefits such as higher incomes, shorter planting cycles, and market opportunities, yet is also shaped by external stakeholders and cultural ties to rice farming. However, the role of stakeholder interactions and socio-cultural norms in shaping farmers' decisions remains understudied. This study aims to (1) identify key stakeholders, (2) analyze how their dynamics influence transition decisions, and (3) assess long-term outcomes for farmers. We employed a decision analysis approach integrating quantitative probabilistic modelling with qualitative methods. Data were gathered through a literature review and semi-structured interviews with 34 individuals in Sinjai District, South Sulawesi, Indonesia. Stakeholders were identified and categorized based on attributes such as experience, availability, relevance, and influence. We developed an impact pathway to explore how stakeholder dynamics affect decision-making and a conceptual model to simulate the expected costs-benefits, and long-term outcomes for farmers. We identified 26 stakeholder types involved in the transition decision, 11 of whom were categorized as potential core experts, most being different types of farmers. Although vegetable farming shows higher long-term income potential, several farmers expressed concerns about food security and the cultural significance of rice, influencing their continued cultivation of rice. This study highlights that economic potential alone does not drive crop transition. Stakeholder relationships and cultural attachments can influence farmers' decision-making. Strengthening collaboration among stakeholders will be essential to support sustainable and resilient agricultural transitions in Indonesia.

Keywords: agricultural transition; cultural value; decision analysis; horticulture; rice farming; stakeholder

Acknowledgement: This study was supported by the German Academic Exchange Service (DAAD) through a scholarship awarded to the first author.